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```
clc;
clear;
close all;
```

===== PARAMETERS

=====

```
M = 16;                % 16-QAM
k = log2(M);

Nfft = 64;             % OFDM subcarriers
Nofdm = 100;           % Number of OFDM symbols
CP = 16;               % Cyclic Prefix
```

===== TRANSMITTER

=====

```
Nbits = Nofdm * Nfft * k;

% Generate random bits
txBits = randi([0 1], Nbits, 1);

% 16-QAM modulation
txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);

% Arrange into OFDM symbols
txSymbolsMatrix = reshape(txSymbols, Nfft, Nofdm);

% IFFT
txOFDM = ifft(txSymbolsMatrix, Nfft);
```

===== CYCLIC PREFIX =====

```
cyclicPrefix = txOFDM(end-CP+1:end, :);  
txOFDM_CP = [cyclicPrefix; txOFDM];
```

===== MULTIPATH CHANNEL =====

```
% Multipath channel coefficients  
h = [1 0.5 0.3];  
  
% Output signal  
rxOFDM_CP = zeros(size(txOFDM_CP));  
  
for n = 1:Nofdm  
  
    % Pass each OFDM symbol through multipath channel  
    rxOFDM_CP(:,n) = filter(h, 1, txOFDM_CP(:,n));  
  
end
```

===== RECEIVER =====

```
% Remove cyclic prefix  
rxOFDM = rxOFDM_CP(CP+1:end, :);  
  
% FFT  
rxSymbolsMatrix = fft(rxOFDM, Nfft);
```

===== EQUALIZATION =====

```
% Frequency response of the channel  
H = fft([h zeros(1, Nfft-length(h))], Nfft).';  
  
% Equalize the received subcarriers  
rxSymbolsMatrix = rxSymbolsMatrix ./ H;  
  
% Convert to serial  
rxSymbols = rxSymbolsMatrix(:);
```

===== DEMODULATION

=====

```
rxBits = qamdemod(rxSymbols, M, ...  
    'OutputType', 'bit', ...  
    'UnitAveragePower', true);
```

===== BER =====

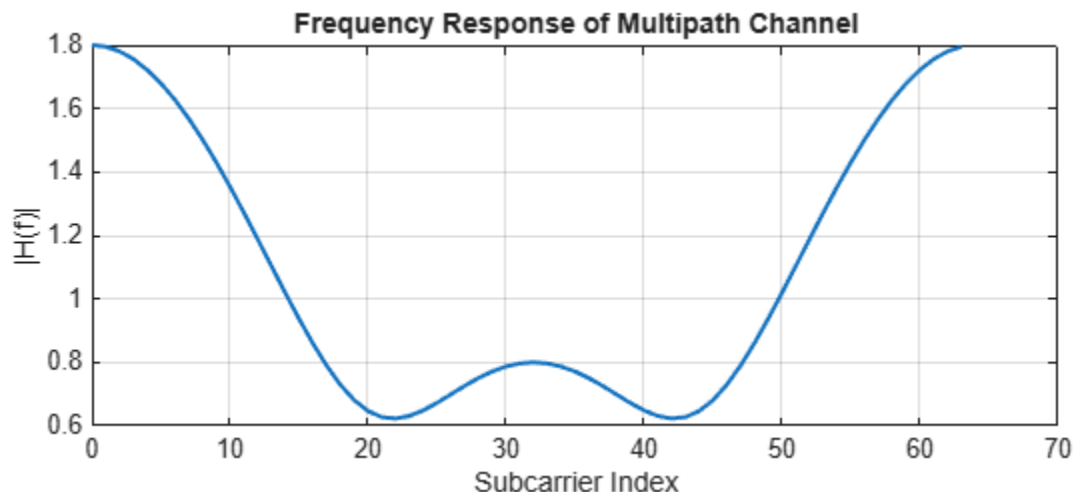
```
[numErrors, ber] = biterr(txBits, rxBits);  
  
fprintf('Number of bit errors = %d\n', numErrors);  
fprintf('BER = %e\n', ber);
```

```
Number of bit errors = 0  
BER = 0.000000e+00
```

===== CHANNEL RESPONSE

=====

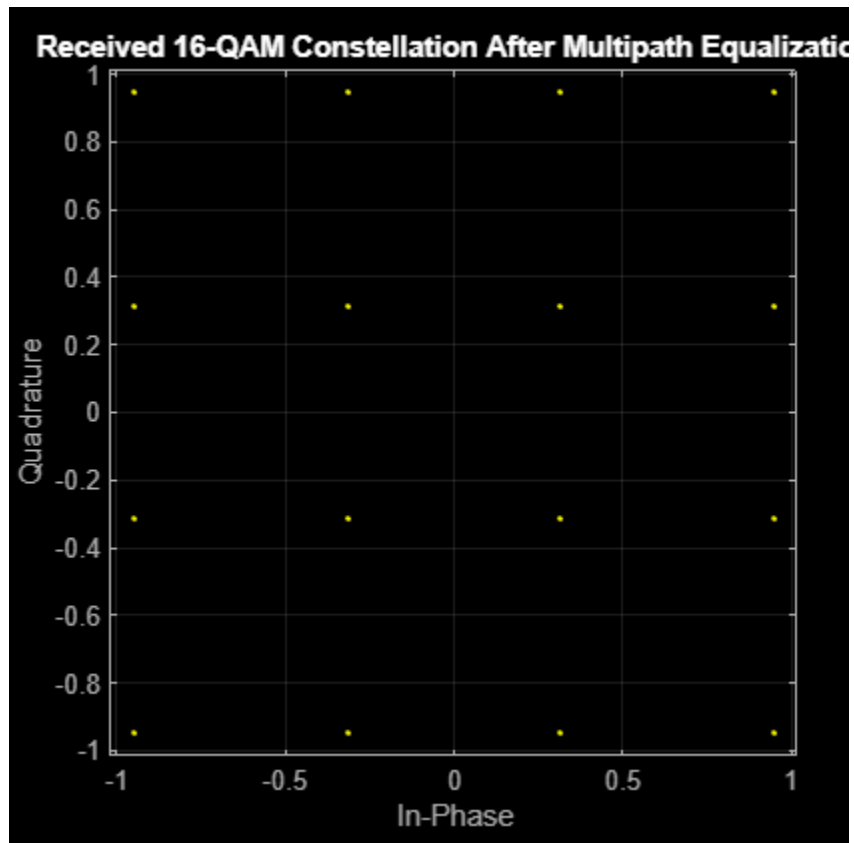
```
figure;  
  
freq = 0:Nfft-1;  
  
plot(freq, abs(H), 'LineWidth', 1.5);  
  
xlabel('Subcarrier Index');  
ylabel('|H(f)|');  
  
title('Frequency Response of Multipath Channel');  
  
grid on;
```



===== CONSTITUTION

=====

```
figure;  
scatterplot(rxSymbols);  
title('Received 16-QAM Constellation After Multipath Equalization');  
grid on;
```



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